Introduction to derivatives, light and energy

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1 Derivatives of polynomials

Note that the derivative of a constant is zero, as a rule. Let a, b be real numbers.

1. Expand $(ax + b)^2$
2. Calculate the derivate of the result.
3. Simplify your answer by removing common factors from the terms.
4. Take the derivate of $ax + b$
5. Compare your results from 4 and 3. Can you guess what the rule is for taking derivatives of brackets?

2 Graphing

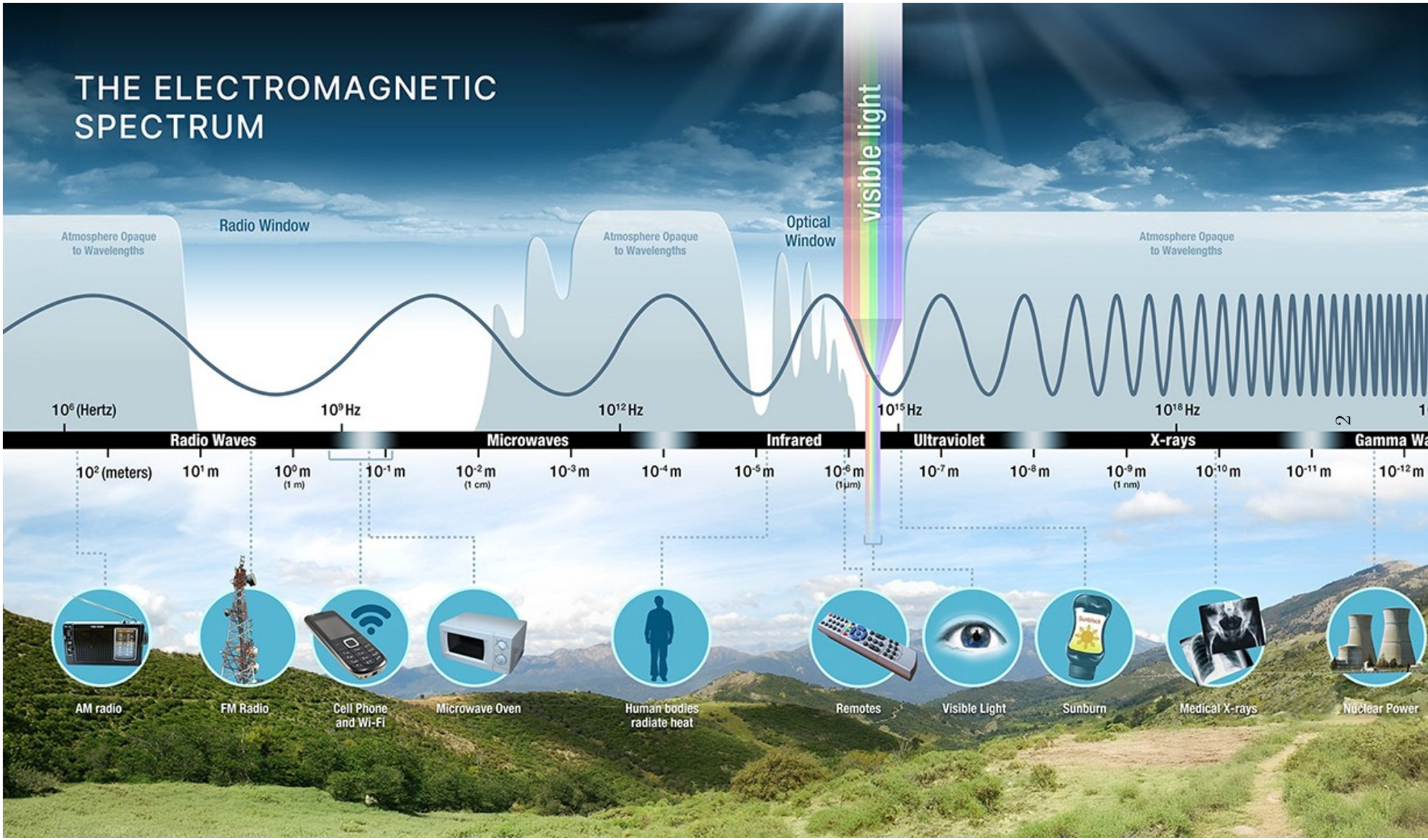
1. Sketch the graph of the function $f(x) = (x^2 - 1)(x + 1)$. Note the location of the roots.
2. Expand the function and take the derivative of the expansion.
3. Find the local minima and maxima by letting the derivative equal to zero and solving for x. Do the results line up with your sketch?

3 Energy, light, and matter

The 'absorption wavelength' (meaning the wavelength of light that would be absorbed by water, is 2.87 micrometres. Note the image below of the electromagnetic spectrum.

Where does 2.87 micrometres appear on the spectrum?

The typical absorption wavelength of glass is 320nm. Where does 320nm appear on the spectrum? What type of light? A microwave oven emits light at a wavelength of 12.2cm. Explain why a glass bowl filled with water would heat the water much faster than the glass itself. This is why you can remove hot food from the microwave without burning yourself (to a point).


THE ELECTROMAGNETIC SPECTRUM
Atmosphere Opaque to Wavelengths
Radio Window
Atmosphere Opaque to Wavelengths
Optical Window
visible light
Atmosphere Opaque to Wavelengths
10⁶ (Hertz)
10⁹ Hz
10¹² Hz
10¹⁵ Hz
10¹⁸ Hz
Radio Waves
Microwaves
Infrared
Ultraviolet
X-rays
Gamma Wa
10² (meters)
10¹ m
10⁰ m (1 m)
10⁻¹ m
10⁻² m (1 cm)
10⁻³ m
10⁻⁴ m
10⁻⁵ m
10⁻⁶ m (1 µm)
10⁻⁷ m
10⁻⁸ m
10⁻⁹ m (1 nm)
10⁻¹⁰ m
10⁻¹¹ m
10⁻¹² m
AM radio
FM Radio
Cell Phone and Wi-Fi
Microwave Oven
Human bodies radiate heat
Remotes
Visible Light
Sunburn
Medical X-rays
Nuclear Power